

The Relationship Between Economic Inequalities and Climate Change

Andy Setser
Adams State University

All people have their beliefs about climate change and the will to do what they can to help ease its effects. Activists have begun biking to places to reduce carbon emissions, eating vegan to reduce greenhouse gasses, recycling to save energy and turning down thermostats to reduce household carbon footprints. The entirety of this upheaval has spurred significant economic research, with consensus among economists that climate change poses an imminent threat and can be effectively addressed through economic perspectives. One of the bigger reports came from economist Nicholas Stern who focussed on the impact that cutting carbon emissions would have on the economy in the short and long run. The primary solution to slow climate change presented by Stern is considered a “climate-policy ramp” that is focused primarily around reducing emissions where the reductions would gradually get tighter. Stern quantified the cost that carbon emissions would have on the future economy and the cost-benefit analysis of the reductions. Stern’s model calculated that a 5 degree celsius (41 degree Fahrenheit) average global warming increase would lead to a 5% decrease to GDP considering only market impacts and even more significant when factoring market externalities (2006). Stern finds that to avoid the damage from global warming that there would have to be a stabilization around 430 parts per million to 550 parts per million of carbon dioxide which would cost 1% of gross domestic product annually and attempting to stabilize sooner would cost a greater loss to GDP (3-4 times more). However, Stern emphasizes the cost of delaying too much which would lead to a higher carbon dioxide stabilization point that would potentially be more costly to fix.

Despite calculating the cost growing as the stabilization is delayed, Stern's model focuses more on the future with a gradual increase of climate related policies as the damages done become more severe in relation to corporate outputs. The solution to climate change presented

by Stern is a viable one, he suggests an optimal carbon price of \$85 per ton which would help balance out the cost of reducing emissions with the benefits of growing climate damages (2006). “The current central estimate of the social cost of carbon is over \$50 per ton in today’s dollars” (EDF, 2017). By following Stern's recommendation and increasing this cost to \$85 per ton, it would incentivize a reduction in carbon emissions as businesses would be more motivated to adopt cleaner practices and technologies to avoid higher cost with the taxation of carbon. The primary critique of this approach is that it only addresses one factor towards global warming. The World Wildlife Fund predicts “Forest loss and damage is the cause of around 10% of global warming” (The Effects of Deforestation). However, Stern's approach does not include solutions other than those related to carbon emissions. Climate change is a complex issue influenced by multiple factors, including deforestation; by ignoring the key contributors in addition to carbon emissions could lead to an incomplete or skewed assessment. The other criticism of Stern’s method comes from the low discount rate he used to calculate the present value of the future cost and benefits to stabilizing carbon emissions. “A zero time discount rate means that future generations into the indefinite future are treated symmetrically with present generations; a positive time discount rate means that the welfare of future generations is reduced or “discounted” compared to nearer generations” (Nordhaus, 2007). William Nordhaus’s review critiques Stern’s 1.4% discount rate used in the analysis stating: “this magnifies impacts in the distant future and rationalizes deep cuts in emissions and indeed in all consumption, today” (2007). A counter discount rate calculated by the UK Treasury’s Green book used a greater variables in the equation for the discount rate with the constant variable (percentage that humanity will not exist in the future) at 1.5% and the rate of change variable (growth per capita income) at 2% which is much higher than Sterns; the change in variables gave the Green Book a

discount rate of 3.5% which seems much more realistic than the rate used in Stern's analysis (Ackerman, 2007). An additional argument by Partha Dasgupta focuses on how the discount rate should represent how different wealth levels are affected from acting on climate change sooner than later (Ackerman, 2007). Since the per capita income is expected to increase year after year despite climate change, the future generations will be richer and the present generation is thought of as poor, and so the discount rate equation would use a greater variable that would lead towards a rate more focused on the present than the future. The method used in Stern's analysis and suggestion to tax higher on carbon emissions is a practical solution that most would agree would work at combating climate change, however, the controversy arises when considering how he calculated the tax that should be put upon carbon and the rate at which these policies should be implemented.

Economists Andrew Schein and Lawrence Goulder write "Carbon Tax V Cap and Trade" which can be read as an additional study for economic approaches towards addressing climate change. The passage compares carbon taxes and cap and trade system policies in the context of greenhouse gas emissions reduction. A carbon tax is set by an external authority and assigns a price to every unit of carbon emitted. A cap and trade involves a limit on the overall permissible emissions and assigning allowances that represent the authorization to emit a specific volume of greenhouse gases. The main advantages to a carbon tax provided in the analysis is the percentage of emission leakage. "Carbon leakage refers to a situation where a company decides to move their production from a country with stringent policies, to a country that is more lenient, leading to an increase in greenhouse gas emissions" (Clear Center, 2020). Goulder and Schein underscore the risk associated with cap and trade, emphasizing that companies engaging in market trading, particularly with oil-exporting countries, may aim to accumulate more carbon at

a lower cost. However, under carbon taxes, even if a country enforces stricter regulations or offers incentives for cleaner technologies, the tax rate remains constant. Carbon taxes are proven to be advantageous again in the analysis as the economist points out that the taxes would go to the federal government's revenue allowing for better funding of public services and programs and possibly forcing a "revenue-recycling benefit" where the taxes collected from carbon could reduce the rates of existing taxes such as income or sales tax. Even with the argument presented that a revenue-recycling benefit would be difficult in the US due to the "carbon tax falling under the domain of the House Ways and Means Committee and Senate Finance Committee...[where the income would be] predisposed toward using the revenues to finance environmental projects and be less inclined toward recycling the revenues" (Goulder and Schein, 2013). The drawback to the revenue recycling benefit doesn't seem overly concerning, though; taxing carbon is already a benefit to climate change itself as it incentivizes against using it, but it will also lead the financial charge to better fund new technology and clean up acts against this pollutant.

Cap and trade systems which are historically more politically favorable have their benefits as well which the article provides. Cap and trade can also provide a greater cost effective approach as it allows more flexibility in how companies achieve their emission reductions. In addition, the linkages across jurisdictions can enhance market liquidity and lead to a more unified effort of combating climate change. In the comparison between carbon tax and cap and trade, there is a clear favorable policy. A carbon tax proves advantageous precisely because it mitigates the flexibility that capitalists might exploit with their profit-oriented mindset. According to Schein and Goulder this includes the elimination of emission leakage and wealth transfers to oil exporting countries, and the benefit of tax reform (2013). Comparing the Schein-Goulder economic analysis and the Stern, a carbon tax stands out as the favorable option.

The main complication comes with trying to figure out how big of a tax and when to implement it; various factors such as time preferences, opportunity costs, inflation, and the considerations of future and present generations well-being. Despite the complexities, the U.S cannot afford not to maintain some form of carbon tax, as it remains essential and economically necessary.

Economist James Boyse goes after a more unique argument that brings economic inequality into the equation of things that cause climate change. Boyse believes the environmental degradation is a direct cause of the power inequalities faced in society (1994). It's no secret that the poor neighborhoods have mitigated the unfortunate things that the better off areas of town don't want, for example, researchers studying toxic waste depots through the U.S. found "a consistent pattern over a 30-year period of placing hazardous waste facilities in neighborhoods where poor people and people of color live." (University of Michigan, 2022). The way this relates to climate change and the degradation of the environment is a new and compelling case that stands out from the traditional climate change arguments. Climate change is one of the most widely debated and contentious topics on a global scale. This discussion fills social media, emerges in celebrity interviews, and takes center stage in political debates. However, amid these conversations, the impoverished class remains notably silent. The undeniable reality is that an individual's voice dwindles in influence when faced with limited power and according to Boyce, "Relative power is a function of three factors: (a) a vector of power-relevant individual characteristics such as wealth, gender, ethnicity, and class; (b) the numbers of individuals involved ; and (c) the political framework in which these variables are mapped to power" (1994).

A cost-benefit analysis is used to add reasoning to why inequality is diminishing the environment. This rationale would imply that an environmentally damaging activity can persist

as long as its incremental net influence on society remains favorable. However, the difficulty arises when considering who the benefit is received by. Due to power inequalities, a cost-benefit analysis might justify an environmentally degrading decision based on significant personal benefits and minimal costs for those in positions of influence. Yet, what unfolds when we factor in the costs imposed on the less privileged? The analysis and argument would achieve greater equilibrium if everyone held the same level of power; the repercussions of decisions couldn't be disproportionately shouldered by underprivileged neighborhoods with limited influence, causing the benefits to diminish. This approach would yield a more comprehensive solution and analysis, ensuring that the challenges are shared by all classes rather than conveniently relegated to the poorer areas of town, unseen by those in positions of power. An additional factor contributing to inequalities leading to a degraded environment is the ability to force change. Even if the more unfortunate communities wanted to enforce a change, "Willingness to pay is constrained of course, by ability to pay" (1994). The willingness for a poor person to pay to improve air quality is less than a rich person's willingness. This isn't indicative of a lack of belief in the importance of enhanced air quality and overall health; rather, it stems from the financial limitations that prevent them from affording such luxuries. To conclude this hypothesis, Boyce touches on the time preference with the benefits and cost in the decision making with the environment. The benefits earned by degrading the environment are short lived while the costs are long term; although the time preference will vary depending on factors like what resource is being degraded and who is receiving the benefit. When considering disparities, individuals with greater wealth tend to exhibit higher time preferences, reflecting less concern for the future. This approach advocates for addressing inequalities and urges those in positions of power to conduct a more comprehensive cost-benefit analysis. This intricate approach, while more complex due to the

multitude of factors influencing the solution to inequality, could potentially yield benefits as a supplementary strategy for addressing global climate change. The proposed strategies in earlier analyses, such as implementing a carbon tax and capping carbon emissions, appear more straightforward. Given the urgency of climate change, I recommend that action be taken in line with the recommendations from all these analyses.

References:

- Ackerman. (2007). Debating climate economics: The stern review vs. its critics.
<https://www.bu.edu/eci/files/2019/06/SternDebateReport.pdf>
- Boyce, J. K. (1994). Inequality as a cause of environmental degradation. *Ecological Economics*.
<https://www.sciencedirect.com/science/article/pii/0921800994901988>
- Clear Center. (2020). What is carbon leakage? . CLEAR Center.
<https://clear.ucdavis.edu/news/what-carbon-leakage>
- EFD. (2017, March 23). The true cost of Carbon Pollution. Environmental Defense Fund.
<https://www.edf.org/true-cost-carbon-pollution>
- The Effects of Deforestation . (n.d.). Learn the effects of deforestation. WWF.
<https://www.wwf.org.uk/learn/effects-of/deforestation#:~:text=Trees%20absorb%20and%20store%20carbon,we%20don't%20stop%20deforestation.>
- Goulder, L. H., & Schein, A. (2013, August 16). Carbon taxes vs. cap and trade: A critical review. NBER. <https://www.nber.org/papers/w19338>
- Nordhaus, W. D. (2007). A review of the stern review on the economics of climate change. *Journal of Economic Literature*.
<https://www.aeaweb.org/articles?id=10.1257%2Fjel.45.3.686>
- Stern, N. (2006, October 30). The Economics of Climate Change: The stern review. Grantham Research Institute on climate change and the environment.
<https://www.lse.ac.uk/GranthamInstitute/publication/the-economics-of-climate-change-the-stern-review/>
- University of Michigan. (2022, November 17). Targeting minority, low-income neighborhoods for hazardous waste sites. University of Michigan News.

<https://news.umich.edu/targeting-minority-low-income-neighborhoods-for-hazardous-waste-sites>

/